

Exp C: does the model’s pre-decoding entropy track its own decoder β ?

May 13, 2026

1 Question

The project proposal hypothesizes (`decoding-modeling.tex`): *language models form a belief about their own decoding parameters by conditioning on the text they have generated*. This experiment tests that hypothesis directly: holding the prompt fixed, vary β_{dec} during self-generation and ask whether the model’s pre-decoding entropy $H(\text{softmax}(\ell_t))$ at late steps reflects the decoder β_{dec} in use.

2 Setup

Qwen-2.5-3B base, 32 prompts (`src/decoding_decoding/prompts.py`), each truncated to its first 10 tokens. Generate 150 tokens per prompt at three β_{dec} settings:

$$\beta_{\text{dec}} \in \{0.5 \ (T = 2.0), \ 1.0 \ (T = 1.0), \ 2.0 \ (T = 0.5)\}.$$

At each step, record $H(\text{softmax}(\ell_t))$ *before* the decoder applies β_{dec} . Late-window aggregation: per-prompt mean H over steps 80–149.

3 First-look result

Table 1: Late-window (steps 80–149) pre-decoding entropy per β_{dec} , with per-prompt paired difference.

condition	late-window H mean (nats) $\pm$ SE	paired diff vs $\beta = 2.0$
$\beta_{\text{dec}} = 0.5$	9.216 ± 0.052	$+8.045 \pm 0.115$ ($t = +69.7$, $n_+ = 32/32$)
$\beta_{\text{dec}} = 1.0$	2.168 ± 0.186	$+1.00$
$\beta_{\text{dec}} = 2.0$	1.172 ± 0.109	—

The late-window entropy moves ≈ 8 nats (factor ≈ 3000 in effective vocabulary) between $\beta_{\text{dec}} = 0.5$ and $\beta_{\text{dec}} = 2.0$. Every one of 32 prompts orders the same way.

Three observations before we believe the headline.

1. Natural high-entropy human writing (WritingPrompts/natural) sits at $H \approx 2.85$ nats — nowhere near the 9.2 nats seen here. So this is not the model ”reading off high-T-like content”; this is a regime that only off-distribution self-sampling reaches.

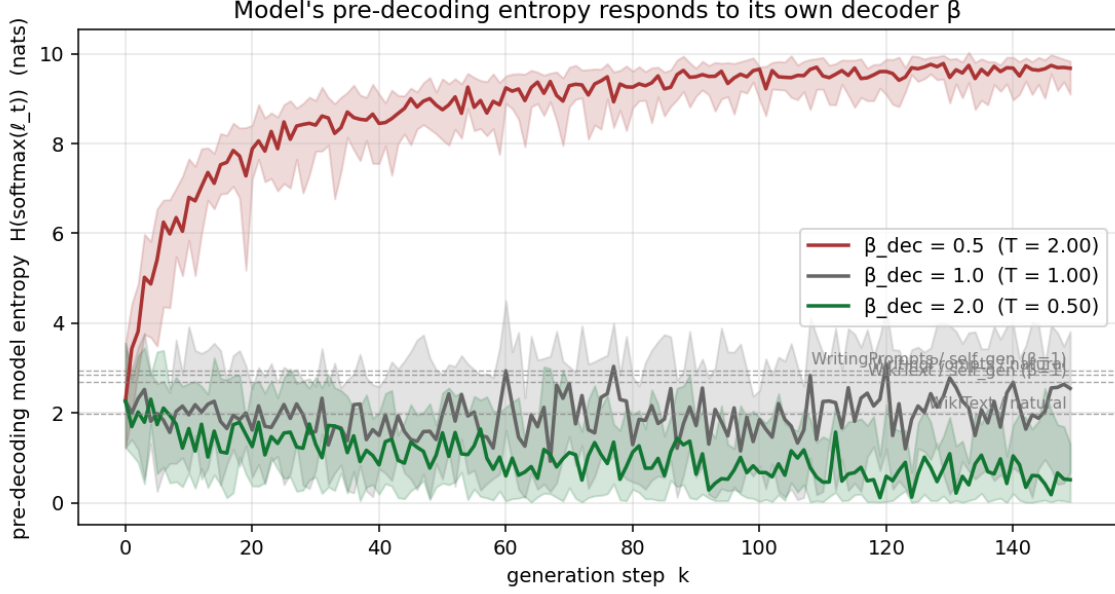


Figure 1: Median $H(t)$ per condition, IQR shaded. Dashed grey lines mark mean entropies of real natural text from the earlier natural_text experiment (Wikipedia ≈ 1.98 , WritingPrompts ≈ 2.85 nats).

2. Inspection of $\beta = 0.5$ generations (Figure 2) shows pure token salad: code keywords, foreign-language fragments, URL slugs, random named entities. $\beta = 1.0$ generations are coherent English. $\beta = 2.0$ generations are also coherent English, more compressed toward common factual phrasing.
3. The entropy trajectory at $\beta = 0.5$ climbs monotonically from ≈ 2 at step 0 to ≈ 9 at step 80: classic compounding off-distribution drift, where each random token makes the next context less parseable and the next sample yet more random.

These three points raise an obvious alternative: the 8-nat effect is *garbage-in, garbage-out* — the model is uncertain because its context is incoherent, not because it has formed a “belief” that the decoder is set to high T .

4 Swap-prefix control

To disambiguate, we hold the early prefix’s generation regime fixed and *switch* the sampler β at step 80. If the model truly has a belief about β_{dec} that updates on recent tokens, then switching to $\beta = 1.0$ at step 80 should pull late-window H down to the $\beta_{\text{dec}} = 1.0$ baseline (≈ 2 nats) within ~ 20 tokens. Five conditions, same 32 prompts:

- pre05_post05 — $\beta = 0.5$ throughout (no switch, control).
- pre05_post10 — $\beta = 0.5 \rightarrow 1.0$ at step 80. **This is the decisive comparison.**
- pre05_post20 — $\beta = 0.5 \rightarrow 2.0$ at step 80.
- pre20_post05 — $\beta = 2.0 \rightarrow 0.5$ at step 80 (mirror).

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 $\beta_{\text{dec}} = 0.5$  (T = 2.00):
«Marie Curie was a Polish-born physicist and chemist whose pioneering research on» a very اءءءء spent $ ;\n Crefit0reement.ToInt_Begin(' About website flood
entirety CommonTouchGMT not[] institution going Bing0000 [][] []stral_debugignon>;\n \n inundlletcherYZ merchant directory genis_BooI Jasmine ast_operations BSP
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 $\beta_{\text{dec}} = 1.0$  (T = 1.00):
«Marie Curie was a Polish-born physicist and chemist whose pioneering research on» chemist who made significant contributions to the study of radioactivity.
Although research is ongoing, it is said that at least 60 percent of the total alpha decay energy of all the potassium 40 in the human body (there are 15 Bq/Kg
potassium 40),

 $\beta_{\text{dec}} = 2.0$  (T = 0.50):
«Marie Curie was a Polish-born physicist and chemist whose pioneering research on» chemist who was the first woman to win a Nobel Prize, and the first person
to win two Nobel Prizes, one in physics and the other in chemistry. She was born on November 7, 1867 in Warsaw, Poland. Her father, Wladyslaw 0ls

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Figure 2: First 60 sampled tokens for prompt 0 (Marie Curie) at each β_{dec} . $\beta = 0.5$ generation degrades to token salad within ≈ 20 steps.

- **pre20_post20** — $\beta = 2.0$ throughout (control).

Table 2: Pre-switch median H and post-switch median H at step offsets from the switch.

condition	$\beta_{\text{pre}} \rightarrow \beta_{\text{post}}$	pre-med	+0	+5	+10	+20	+50
pre05_post05	0.5 $\rightarrow$ 0.5 (no switch)	8.42	9.26	9.21	9.49	9.66	9.47
pre05_post10	0.5 $\rightarrow$ 1.0	8.56	9.44	9.10	9.10	8.90	8.92
pre05_post20	0.5 $\rightarrow$ 2.0	8.57	9.53	6.83	4.33	1.68	0.33
pre20_post05	2.0 $\rightarrow$ 0.5	1.21	0.69	5.20	6.46	8.01	9.06
pre20_post20	2.0 $\rightarrow$ 2.0 (no switch)	1.29	0.64	0.58	1.05	0.30	0.39

What this shows.

- **pre05_post10**: switching the sampler from $\beta = 0.5$ to $\beta = 1.0$ leaves median H essentially unchanged (8.56 $\rightarrow$ 8.92 at step +50). The model does *not* update its predictions to match the new sampling regime over even 50 tokens of new evidence.
- **pre05_post20**: switching to $\beta = 2.0$ *does* recover coherence over ~ 20 tokens. But this recovery is explained by the mechanism of $\beta = 2$ sampling itself — $\beta = 2$ forces a top-1 selection from $\text{softmax}(\beta \cdot \ell_t)$, which under a flat ℓ_t still picks the highest-probability token (typically a common safe continuation like punctuation, common articles, frequent function words); those tokens then *re-anchor* the context to in-distribution English. This is not the model "updating its belief"; it is the sampler injecting in-distribution tokens that gradually overwrite the bad prefix.

The decisive comparison **pre05_post10** falsifies a strong form of the "decoder-belief" framing. If model-belief-about- β were a fast posterior update on recent sampled tokens, switching the sampler should be visible within a handful of tokens. It is not.

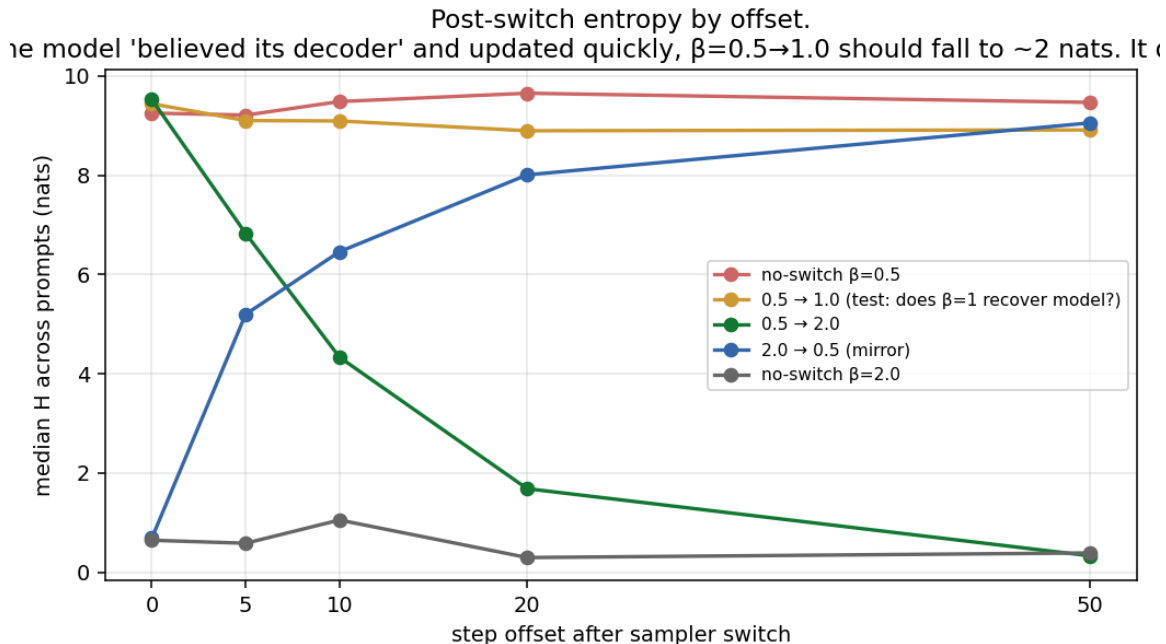


Figure 3: Median pre-decoding H at offsets $+0, +5, +10, +20, +50$ after the sampler switch. Critical observation: *the orange line (pre05_post10) does not fall* — the model does not recover coherence when the sampler switches to $\beta = 1.0$.

5 What survives

A weak form of the framing remains testable and may still be true, but is not established by this experiment:

- At $\beta_{\text{dec}} = 2.0$ self-generation, late-window $H \approx 1.17$ nats — *lower* than the model’s entropy on real WikiText (1.98 nats from `natural.text/macros.tex`). This ≈ 0.8 -nat gap is consistent with the model sharpening below what the natural content alone would justify, when its prefix matches the $\beta = 2$ regime. But it is also consistent with $\beta = 2$ generation producing especially predictable content (top-1 sampling produces high-frequency tokens), so this gap is suggestive, not decisive.
- The earlier natural-text experiment found a small calibration offset on real creative writing ($\Delta \log \hat{\beta} \approx -0.24$, $t = -3.8$). That offset, on real human text, is a much weaker but cleaner signal than the Exp C numbers and is the surviving load-bearing positive evidence for the project’s framing.

6 Bottom line

- The 8-nat entropy spread between $\beta_{\text{dec}} = 0.5$ and $\beta_{\text{dec}} = 2.0$ self-generation is mostly the model *losing coherence* as its own off-distribution sampling decoheres the context. It is not a posterior on β_{dec} .
- The swap-prefix control confirms this: switching the sampler from $\beta = 0.5$ to $\beta = 1.0$ does not pull H back to the $\beta = 1.0$ baseline over 50 tokens. A model with a fast posterior on its

decoder would recover.

- The project’s framing must scope down: the model’s distribution does not aggressively update on the apparent decoder of its own generated prefix at the timescales tested here. Smaller signals — e.g. the WritingPrompts/natural offset from the natural_text experiment, or the ~ 0.8 -nat sharpening below WikiText at $\beta = 2.0$ — are the remaining candidates for clean “decoder-belief” evidence and need a different experimental design to isolate.